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The impact of body surface convex angle on dressed waist ease

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Abstract

Purpose – The comfort and fit of clothes are affected by fabric properties, dressed ease and environmental conditions, in which dressed ease is influenced by the interaction among complex shapes of human body, style design and fabric mechanical properties.

Design/methodology/approach – In this study, the dressed ease distribution at waist section, which is related to body surface convex angle, was investigated using 3D scanning. A series of surface convex angles on bust and back were formed after adjusting the mannequin. The mannequin was scanned by TC2 separately in garments with eight different ease allowances. Then the dressed ease distributions at waist under different convex angles of body surface have been acquired by calculating the distance between waist points and dressed surfaces along normal directions.

Findings – The results showed that the body surface convex angle was weakly related to the dressed ease when the garments' bust ease allowance was below 4 cm. When the garments' bust ease allowance was within 6–12 cm, the body convex angle had a great impact on the dressed waist ease distribution in the condition of 26º–33º bust convex angle and 13.96º–17.96º back slope angle. For slack garments with more than 16 cm ease allowance, the dressed waist ease distribution did not relate to the bust convex angle, while it strongly related to the bust convex angle between 13.96º and 17.96º. The regression model was statistically significant between the dressed ease value and the body surface convex angle.

Originality/value – According to the dressed waist ease distribution of different body surface convex angles, this paper gives an application of pattern modification in order to optimize the waist fit. The results can provide guidance for the optimization of different body shapes. At the same time, the application of gap data to 3D virtual fitting can greatly improve the authenticity of virtual simulation effect.

Keywords Dressed ease, Convex angle, Ease allowance, Waist section, Garment pattern

Paper type Research paper

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Introduction

Ease allowance is usually defined by circumferential differences between the garment and body in apparel industry. When garment was worn, ease allowance was converted into dressed ease in a spatial form. Dressed ease can influence body-conforming fit, thermal-moisture comfort and the accuracy of 3D virtual fitting Liu *et al.* (2012). Body-conforming fit is the most concerned issue for people to choose clothes (Ashdown and Loker, 2001; Erwin *et al.*, 1979). Body shape has been increasingly recognized as a fundamental factor that the apparel industry must consider for a good fit because women with identical measurements can have completely different body shapes, resulting in various fit problems Namsoon Kim *et al.* (2018).The essence of personalized pattern design of clothing is to combine human body shape and ease allowance and then shape into a fitting space according to the style. Several large apparel companies, such as Levi's and The Limited, have provided "Curvy" and "Straight" pant styles for customers with different body types (Levi Strauss & Co. Levi's Curve ID, 2016; The Limited. The One Fit, 2016). And department stores such as Nordstrom have provided dress fit guides for various figure types (Nordstrom Incorprated. Women's Dress Fit Guide, 2016). Therefore, the dressed ease and body shape play key roles in pattern design and modification. The contribution of the dressed ease to the clothing fitness and thermal-moisture comfort has been widely investigated. Differences in fabric properties produce differences in garment fit and the basic pattern should be changed (Ashdown and Dunne, 2006). Currently, the majority of the research on dressed ease has been focused on the dressed ease distribution on a manikin through 3D body scanner, including the relationship between dressed ease distance and dressed shape (Xu *et al.*, 2008; Xu and Zhang, 2009; Zhang *et al.*, 2012) and the relationship between fabric mechanical properties and dressed ease shape (Agne Lage and Kristina Ancutiene, 2017; Yu *et al.*, 2012). Simona in his research was to examine the fit of a skirt on a live model and on virtual models such as parametric and scanned body models in order to propose which virtual human body is the most suitable where garment fit is concerned (Simona Jevšnik *et al.*, 2015). Although it is crucial for the comfort and the fit of a garment (Thomassey, 2013), the dressed ease distribution is not easy to be identified and evaluated accurately because of its tiny and complicated shape. In the applied CAD software, the manikins and the dressed model are mainly constructed based on spring particle model and surface collision detection method, which are physical and geometric simulations (Simeon Gill, 2015).However, designers still think that it is hard to set an appropriate ease in garment fitting model. The main reason is that it is quite difficult in setting dressed ease value under different body shapes and various fabrics. Thus, a pattern designer should estimate how garment style is affected by the characteristics of the fabric and the body shapes. Pattern-making methods are under investigation to solve this problem (Cho *et al.*, 2005; Huang *et al.*, 2012).

It can be confirmed that garment forms a variety of shapes based on human body shape. From the analysis of the spatial component of body dressed shape, human body shape plays a basic and vital role in the construction and distribution of dressed ease. The aforementioned scholars mainly studied the influence of garment ease allowance and fabric physical properties on the dressed ease distribution, while the impact of human body shape on the dressed ease distribution has rarely been published.

The aim of this research was to investigate dressed ease distribution at waist section with different convex angles of body surface. Researchers use 3D body scanner ([TC]2 NX-16.US) to collect the data of the mannequin naked and worn. After reconstructing the models by reserve engineering software, the dressed ease distance between the point on the human body surface and the garments along the normal was calculated, and the dressed ease distributions at waist under different convex angles of body surface were analyzed. The relationship between the convex angles and the dressed ease distribution was analyzed under different garments' ease allowance. The regression formulas between the dressed ease at waist section

and the bust convex angle, back slope angle and garments ease allowance were established. According to the dressed ease distribution at waist, researchers adjusted the dividing lines of the pattern to optimize the fit degree. The waist cross sections in validation experiment showed that the dressed ease distribution was modified evenly. It can also be used as a reference for the high-fidelity 3D garment virtual fitting.

Experiments

Given an ease allowance at a waist section, the dressed ease would be distributed around the waist section depending on factors such as the clothes structure, material properties, body size and its posture. In this experiment, a sleeveless *T*-shirt and a female mannequin were chosen for 3D scanning.

Material and methods

The 3D body scanner ([TC]2NX-16.US) can only scan the outermost surface of light-colored garment. The purpose of this paper is to study the relationship between the dressed ease of waist section and the angle of body surface. Relevant studies show that the distribution of the dressed ease between the bust and waist sections was not affected by the mechanical properties of the fabric (Xu *et al.*, 2008; Xu and Zhang, 2009), so the properties of the fabric are not taken into consideration. Light-colored pure cotton fabrics widely used in the garment industry were selected as the experimental sample garment materials, and the fabric properties are shown in Table I. Japanese prototyping method was used to produce eight samples of collarless and sleeveless garments with different size, numbered 1# to 8# (Figure 1). Previous research shown that the ease allowance affected the dressed ease in

Fabric	Fabric structure	Fabric composition	Density (The root/10 cm) Wrap	Weft	Thickness (mm)	Drape coefficient
Cotton fabrics	Plain	100% Cotton	400	340	0.18	46.00

TableI. The fabric performance

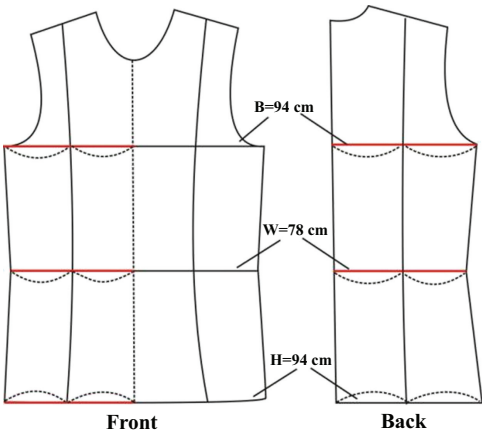

Figure 1. Pattern of the experimental garments

different degree when the bust was given tight ease (0–4cm), fit ease (6–12)cm and loose ease (16–24cm) (Xu *et al.*, 2008; Xu and Zhang, 2009; Zhang *et al.*, 2012). This paper made one tight clothes, four fit clothes with 2cm difference and three loose clothes with 3 cm difference. Waist circumference and hip circumference of tight and fitting garments can be made according to $W = B-16$ and $H = B$, while loose garments can be made according to $W = b-14$ and $H = B$. The garment specifications are shown in Table II.

There are lots of convex angles on human body surface. They influence the specified dressed shape section according to their location. This paper investigated the impact of bust and back shape on waist ease. Therefore, the two surface angles shown as in Figure 2, bust convex angle and back slope angle, which reflect the shape of bust and back, were selected as the research objects.

In order to avoid the error caused by breathing and different poses, we chose mannequins as the objects to measure the ease. According to China Standard GB/ T1335.2-1997, the mannequin of 160/84A specification was selected. On the basis of the original bust convex angle and back slope angle on the mannequin, six various bust convex angles were formed by wearing bust pads with different thicknesses as shown in Plate 1. The mannequins were numbered as X1 to X7. Table III is the detailed specification of the bust convex angle. In the same way, three different back slope angles were made with a deviation of 2° and numbered as Y1 to Y3. Based on the original back slope angle, Table IV presents the parameter specification of the back slope angle. The bust convex angle and back slope angle selected in the experiment covered all the ranges of body types in Chinese females.

Table II. Size of the experimental garments

Position	Body size	1#	2#	3#	4#	5#	6#	7#	8#
B (cm)	84	86	90	92	94	96	100	103	106
$\triangle B$ (cm)	0	2	6	8	10	12	16	19	22
W (cm)	64	70	74	76	78	80	86	89	92
$\triangle W$ (cm)	0	6	10	12	14	16	22	25	28
H (cm)	89	92	96	98	100	102	104	107	110
$\triangle H$ (cm)	0	3	7	9	11	13	15	18	21

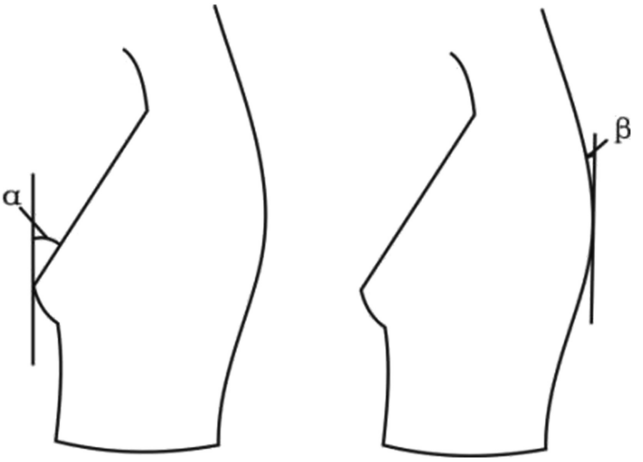

Figure 2. Bust convex angle and back slope angle

Plate 1. Body surface angle setting and measurement

Bust convex angles	X1	X2	X3	X4	X5	X6	X7
Angles degree/(°)	13.96	24.49	26.53	27.93	28.10	31.22	33.42

Table III. Bust convex angle specification

Back slope angles	Y1	Y2	Y3
Angles degree/(°)	13.96	15.96	17.96

Table IV. Back slope angle specification

Test procedure

Researchers adopted [TC]2NX-16 as the experimental equipment.

(1) Determined the position and posture of the mannequin, the cloud points of the undressed mannequin were obtained through $[TC]^2$NX-16.

(2) Fixed the position and posture of the mannequin, garments 1# to 8# were worn for scanning respectively and obtained the cloud points of mannequins with 1# to 8# garment.

(3) Scanned the modified mannequins wearing bust pads to obtain the modified mannequin data. Then garments from # 1 to # 8 were successively inserted for scanning and the cloud points were obtained.

(4) In the same way, the cloud points of eight garments were obtained after transforming the mannequin with different back slope angles and garments.

In order to avoid the human error of experimental operation, each garment was scanned three times under each grade difference specification, and the mean value was taken as the experimental result.

Dressed ease calculation

The cloud points of mannequins were acquired by 3D body scanning. After outputting to reverse engineering software, researchers built the mannequin model, which was edited only to keep the torso, as shown in Plate 2. The model of the dressed mannequin was dealt with the same method. Then, the naked and the dressed torso models were aligned based on the position coordinates in Plate 3. Researchers applied the method of the plan construction to select the waist section required for the research (Wang *et al.*, 2012).The coordinates of 60 points at the waist cross section were collected by the equidistant sampling method. The waist cross-section curve of the naked and the dressed mannequin were fitted. As shown in Figure 3, the distance between the points on the body and the clothes curve was calculated along the normal direction, which was defined as the dressed ease distance at the waist section.

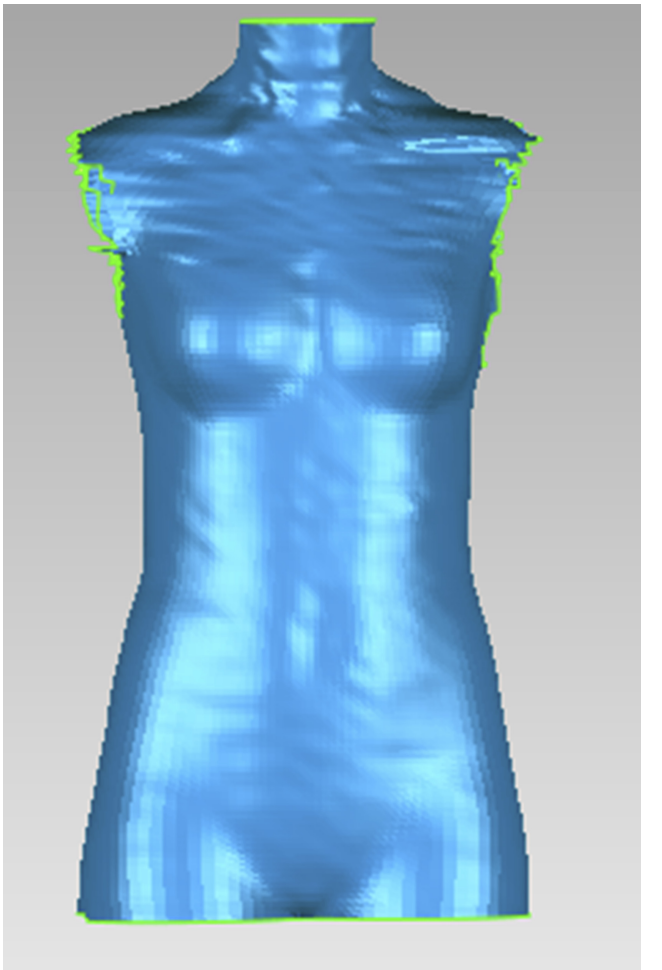

Plate 2.
Manikin torso

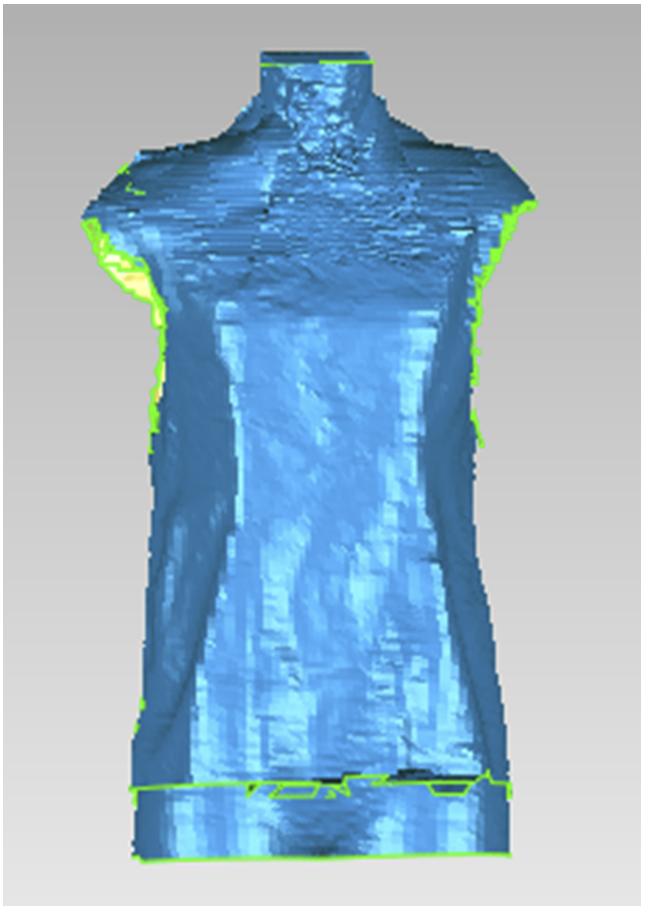

Plate 3.
Alignment of manikin and dressed model

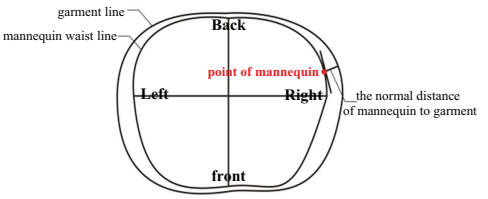

Figure 3.
Dressed ease distance calculation method at waist

Results and discussion

Evaluation of the dressed ease at waist section

In order to illustrate the relationship between the surface convex angles and the ease distribution at the waist section more clearly, the cross section was segmented into 12 regions with every 30° interval angles. The dressed ease values of the cross points at waist cross section and separation lines were calculated under various surface convex angles. As shown in Figures 4 and 5, dressed ease distribution at the waist section from 0° to 180° was similar to

the dressed ease from 180º to 360º. So, this paper only took the regions from 0º to 180º as the objects to study the dressed ease distribution rule in correlation analysis.

Effect of bust convex angle on dressed ease at waist section

Researchers divided the waist cross section into four regions: front, back, left and right side regions. Figure 6 shows that the dressed ease distribution in four regions varied with different bust convex angles. The dressed ease distribution at waist section demonstrated that the ease distance spread variously and unevenly with different bust convex angles at the waist cross section. The greatest influence upon dressed ease at waist section was obtained when experimental garments were 2# to 5#, as shown in Figure 6. It is easy to discern that the influence of bust convex angle on the dressed ease distribution at the waist became smaller while the ease allowance increased. When the clothes number was 8#, the effect of the bust convex angle on amount of the ease distance was the smallest.

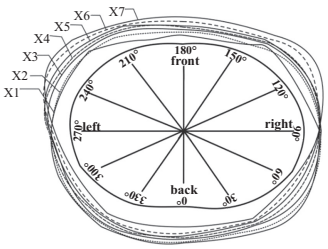

Figure 4.
Dressed ease of different bust convex angles

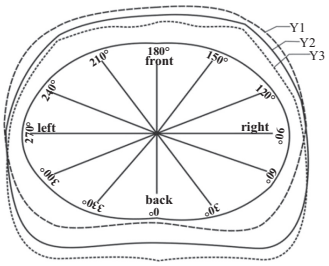

Figure 5.
Dressed ease of different back slope angles

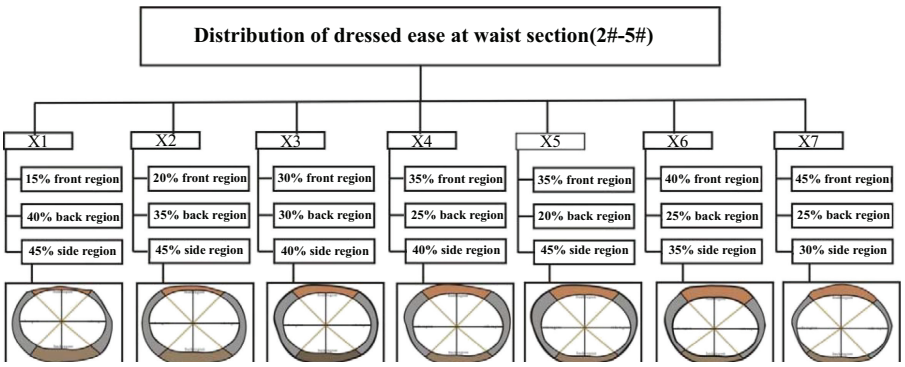

Figure 6.
The dressed ease distribution in four regions at the waist section

As illustrated in the figure, with the increment of the bust convex angle, the dressed ease at waist section gradually shifted from the back regions and side regions to the front regions. When the bust convex angle increases from X1 to X5, the dressed ease at waist section mostly transfers from the back regions to the front regions, while the ease of side regions transferred to the front regions is less. However, when the bust convex angle specification is increased from X6 to X7, the dressed ease at waist section mostly transfers from the side regions to the front regions, while the ease of the back regions hardly transfers to the front regions.

The bust convex angles that influenced the dressed ease distribution weakly were X1 and X2. As illustrated in Figure 7, the dressed ease distribution at the waist with different specification garments is almost the same under X1 and X2 convex angles. Because thoracic angle is small and flatter, the whole bust supported the clothes hardly, thus the influence on the dressed ease distribution at the waist section is relatively weak. To the contrary, X7 has the greatest impact on the ease distance at the waist due to the larger bust convex angle, which leads to the three-dimensional rise of the whole bust of the body and plays a good supporting role for clothes. At this time, the dressed ease distribution with various ease allowances is quite different.

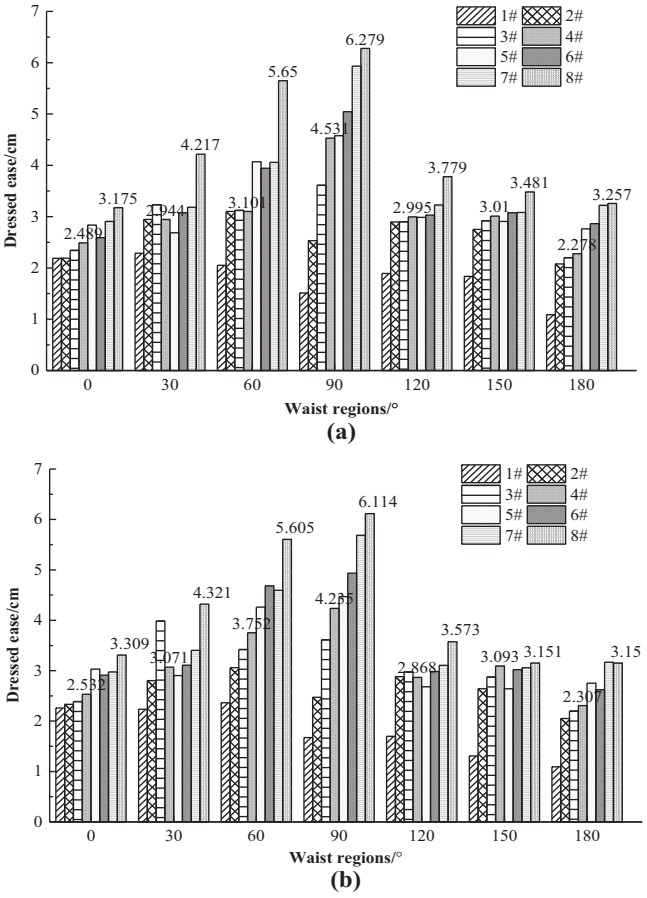

Figure 7. Dressed ease distribution at waist with different garments. (a) X1 bust convex angle specification. (b) X2 bust convex angle specification

The bust convex angles that influenced the dressed ease distribution greatly were X1 to X7 when ease allowance was 10 cm. As shown in Figure 8, the dressed distribution at the waist with same specification garment is different under X1 to X7 convex angles. The fit of garment to human body with different surface convex angle is different. In this study, the dressed ease deviation is selected as the index to measure the fit of garment when wearing same-specification garment; the mathematical expectation of dressed waist ease at different waist regions is selected as the center value of the deviation. The smaller the deviation value is, the more fit the garment has. As illustrated in Table V, the deviation is generally the smallest when the bust convex angle specifications is X4, which is most suitable for the human waist. Meanwhile, the distribution of dressed ease at waist section is shown in Figure 8c. In Figure 8c, dressed ease at waist section is mostly distributed in the side regions of the human body, and the dressed ease at waist section of back regions is more distributed than that in the front regions. Although the bust convex angles became smaller, the dressed ease at waist section was more distributed in the side regions. When the bust convex angles increase, the dressed ease at waist section of waist sections is more distributed in the front regions of the human body. Therefore, the difference of bust convex angle leads to changes in the fit of the garment under the same ease allowance. This proves that using more garment patterns under same ease allowance when body was different could achieve better garment fitting to the body.

Effect of back slope angle on dressed ease at waist section

Figure 9 describes the distribution of dressed ease in waist section at different back slope angles. How does the dressed ease amount of waist section change with the increase of back slope angle? When the back slope angle was Y1, dressed ease at waist section distribution

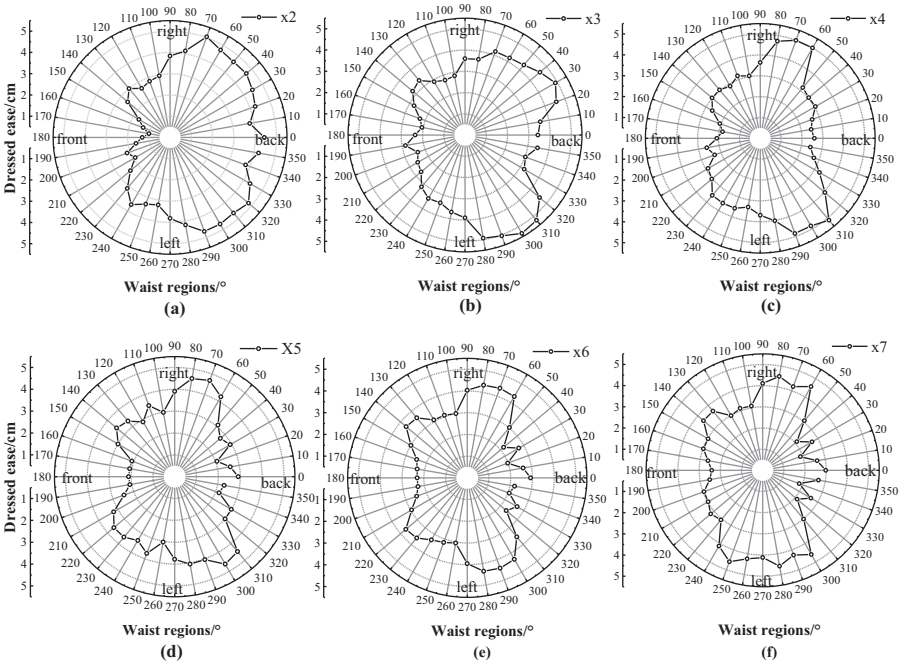

Figure 8.
Dressed ease distribution at waist with same garment

mainly consists of the front regions accounting for 30 percent of the whole regions, the side regions account for 40 percent and the back regions account for 30 percent. With the increase of back slope angle, dressed ease at waist section gradually transfers from the front regions to the back and the side regions, and the trend of transfer became gentler with the increase of back slope angle, as shown Figure 9. Because the distance between the scapula of the body and the waist of the body was relatively long, it had less effect on the dressed waist ease distribution with the increase of back slope angle.

Researchers divided the garments into three groups according to the ease allowance. In the first group, the bust ease allowance was 0–4 cm (garment1#). As can be seen from Figure 10, with the increase of the back slope angle, the dressed ease at the waist section of 60° area changed, while other areas remained unchanged. This indicates that the back slope angle of tight garments has little influence on the distribution of dressed ease of waist section. Because garment 1# was a body-fitting garment, the dressed ease distribution at waist section was less affected by the scapular support point. The bust ease allowances of the second and third groups were 6–12 cm (garments 2# to 5#) and 16–24 cm (garments 6# to 8#). Difference back slope angles have a great impact on the dressed ease distribution at waist section. The dressed ease in the back regions gradually increased and other regions became smaller with the increase of back slope angles, and the most obvious change was in the regions of 0° to 30°. Since the scapula on both sides of the human body supports the garment with a certain amount of ease allowance, the greater the back slope angle, the more prominent the scapula was, which leads to a significant change in the dressed ease at waist section of the back regions.

Waist regions of bust convex angle	X1	X2	X3	X4	X5	X6	X7
0° (cm)	−0.872	1.371	0.366	−0.086	−0.153	−0.15	−0.472
30° (cm)	−0.444	1.083	1.516	−0.344	−0.455	−0.653	−0.702
60° (cm)	0.517	0.159	−0.403	0.026	−0.372	−0.27	−1.408
90° (cm)	−0.274	0.23	−0.208	−0.174	0.103	0.225	0.297
120° (cm)	0.052	−1.125	0.212	0.042	0.154	1.715	0.136
150° (cm)	1.064	−0.995	0.075	−0.036	0.303	0.299	0.288
180° (cm)	−0.076	−0.795	0.241	−0.024	0.606	0.198	0.302

Table V. Deviation of dressed ease

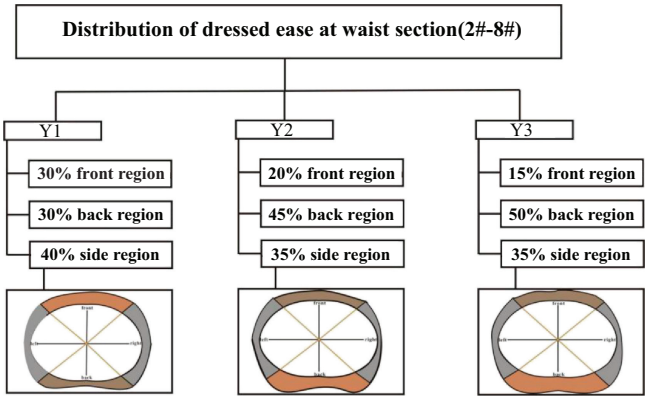

Figure 9. The dressed ease distribution in four regions at the waist section

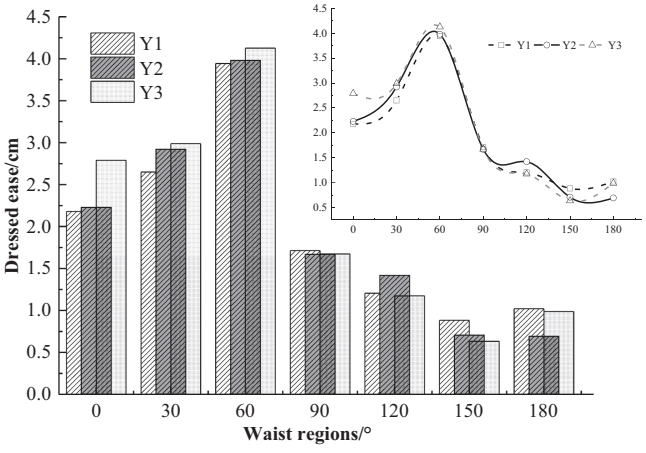

Figure 10.
Dressed ease distribution at waist section with different back slope angles

Table VI.
Correlation analysis of dressed ease and bust convex angles

Waist regions/(°)	Pearson correlation coefficient 1#	2#	3#	4#	5#	6#	7#	8#
0	0.471	0.689*	0.765*	0.692*	0.721*	0.581*	0.457*	0.543*
30	0.412	0.912**	0.927**	0.877*	0.918**	0.519*	0.436	0.323
60	0.431	0.899*	0.874*	0.798*	0.902**	0.621*	0.785*	0.579*
90	0.216	0.135	0.298	0.371	0.234	0.221	0.178	0.157
120	−0.525*	−0.847*	−0.941**	−0.936**	−0.966**	−0.314	−0.353	−0.427
150	−0.527*	−0.938*	−0.935**	−0.911*	−0.899*	−0.412	−0.476	−0.535*
180	−0.314	−0.643*	−0.628*	−0.649*	−0.661*	−0.532	−0.345	−0.217

Note(s): *represents a significant correlation at the 0.05 level; **represents a significant correlation at the 0.01 level

Table VII.
Correlation analysis of dressed ease and back slope angle

Waist regions/(°)	Pearson correlation coefficient 1#	2#	3#	4#	5#	6#	7#	8#
0	−0.443	−0.907**	−0.913**	−0.974*	−0.929*	−0.899*	−0.934**	−0.976*
30	−0.458	−0.968*	−0.903*	−0.911*	−0.989**	−0.924**	−0.957**	−0.903*
60	−0.289	−0.791*	−0.774*	−0.906**	−0.931**	−0.964**	−0.903**	−0.911*
90	0.211	−0.316	−0.327	−0.435	0.176	0.357*	0.311*	0.284*
120	0.399	0.632**	0.799*	0.727*	0.761*	0.723*	0.607**	0.555*
150	0.368	0.874*	0.910**	0.792*	0.832*	0.793*	0.867*	0.822*
180	0.232	0.876*	0.919**	0.833*	0.884*	0.903*	0.856*	0.829*

Note(s): *represents a significant correlation at the 0.05 level; **represents a significant correlation at the 0.01 level

Correlation analysis

Table VI shows the correlation between the dressed ease and bust convex angles at different ease allowance garments: for the garments 2# to 5#, the correlation in the 30° to 60° regions and the 120° to °150 regions reached 0.01, the 180° region and 0° region were moderately

correlated (sig > 0.5), 120° regions has the highest correlation and the correlation coefficient is above 0.9, because this region was located between the side seam and the BP point of the human body, indicating that the data analysis is consistent with the actual situation. 90° region has the lowest correlation, because the arm's restraining effect on the garment results in that the dressed ease at the side seam of the human body is not affected by the change of bust convex angle. For the garments 1#, 6# to 8#, there was a moderate or low correlation between dressed ease at waist regions and bust convex angle. This shows that garment with too much or too little ease allowance is not affected by the bust convex angle.

Table VII shows correlation between dressed ease and back slope angles at different ease allowance garments: for the garments 2# to 8#, correlation significance in the 0° to 60° regions is less than 0.01, 0° to 30° regions is 0.001, correlation coefficient is above 0.9, consistent with the aforementioned analysis. The correlation significance of the 120° to 180° regions is less than 0.05, further indicating that the back slope angle has a significant effect on dressed ease distribution with some ease allowance garment. For the 1# garments, back slope angle has little effect on the distribution of dressed ease, and the dressed ease between 0° and 30° regions has a low correlation with back slope angle. The dressed ease and back slope angle are not correlated at 60° to 180° regions.

Regression analysis

According to the correlation analysis, there was a high correlation between the dressed ease and bust convex angles, back slope angles and ease allowance in 2# to # 5 # garments. In this study, multiple linear regression analysis was performed for the amount of ease in the 0°, 30°, 60°, 120°, 150° and 180° regions, with the values of bust convex angles, back slope angles and ease allowance as variables. The regression results are shown in Table VIII, where y is dressed ease , x is ease allowance, α is bust convex angle and β is back slope angle (before regression analysis, the bust convex angle and back slope angle are converted into radian and angle systems using the formula $\theta^0 = \frac{\pi}{180} \times \theta$).

Table VIII shows the regression equation of dressed ease with bust convex angles, back slope angles and ease allowance. As shown in the table, the regression equations for dressed ease in the 30° and 120° regions are the best, and R^2 values are all above 0.95. 60° region and 150° region have a slightly better goodness of fit, with R^2 values of 0.776 and 0.879.

Application

Fit of garment varied under different body types when worn the same style and ease allowance garment. Waist comfort has an important impact on the fit of garment (Lu, 2014). Therefore, this studied revised the pattern of 2# garment worn by mannequin with different bust convex angles to make it have a better fit in the waist. Based on the dressed ease distribution at different bust convex angles, from the earlier analysis, researchers can see that the deviation value of the human body with X4 specification bust convex angle was the

Waist regions (/°)	The regression equation	R^2
0	$y = 1.019x - 4.31\alpha + 12.13\beta + 0.277$	0.935
30	$y = 1.021x - 6.724\alpha + 8.632\beta - 0.897$	0.968
60	$y = 0.907x - 5.398\alpha + 6.954\beta + 1.12$	0.776
120	$y = 1.371x + 8.76\alpha - 5.381\beta - 0.32$	0.979
150	$y = 0.312x + 6.037\alpha - 3.768\beta - 0.75$	0.879
180	$y = 0.899x + 5.71\alpha - 6.018\beta + 1.03$	0.907

Table VIII. Regression equations for different regions

smallest on the whole when waist ease allowance was 10 cm, which means the waist was most suitable. Therefore, the garments worn with other bust convex angles were modified with dressed ease distribution, Figure 8c as the reference standard, so that the garments with other bust convex angles with a waist allowance ease of 10 cm were fit.

The reason for dressed ease distribution difference in the same specification garment for different bust types of human body was that the dividing line region of garments will change with the change of bust convex angle to some extent, thus affecting the fit of garment. Therefore, this study takes the position of the dividing line with x4 specification bust convex angle as the standard. By changing the position of the dividing line of other garment, the 2# garment can achieve the fitting state under different bust convex angle body. As illustrated in Figure 8a,b, dressed ease at waist section is mainly distributed in the back regions, and most distributed in 0º to 30º regions when bust convex angles are X1 to X3. At this time, the dividing line of garment was moved forward to change the distribution state. Dividing line on the front piece of the garment was offset to the center front seam with a step of 0.25 cm, while dividing line on the back piece was shifted to the side seam with a step of 0.25 cm. When bust convex angles were X4 to X7, dressed waist ease was mostly distributed in the front regions of human body. We can observe in Figure 8d–f, dressed ease at waist section became smaller in center back region when bust convex angle became larger. Dividing line of front piece was shifted to the side seam with a step of 0.25 cm, and dividing line of *t* back piece was shifted to center back seam with a step of 0.25 cm, so that the dressed ease at waist section was evenly distributed as shown in Figure 8c. Variation of dividing line is shown in Figure 11.

The pattern of garment 2# was modified by the aforementioned method, and then the corresponding modified garments were worn on the mannequin with different bust convex angles. The dressed ease at waist section was obtained through aforementioned experimental method. As shown in Figure 12, dressed ease distribution at waist section of mannequin with different bust convex angle was relatively uniform after the revised garments were worn, and they were all closed to the dressed waist ease distribution under the reference standard. Deviation value of different bust convex angle at waist regions was small and the difference between them was small, as showed in Table IX. Plate 4 shows the virtual fitting effect before and after the modification of the garment pattern. It can be seen that after the modification of the garment version, the fit of the garment was significantly improved in the waist and abdominal bulge of the virtual human body. This conclusion indicated that the modification of the pattern was successful.

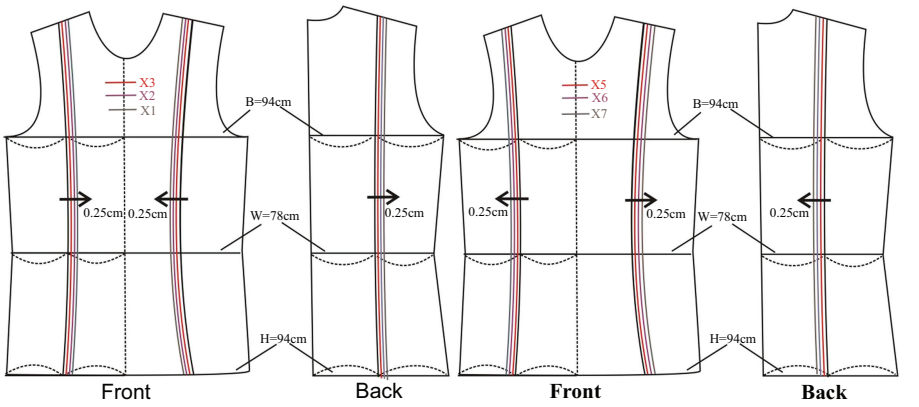

Figure 11.
Pattern of modified garments

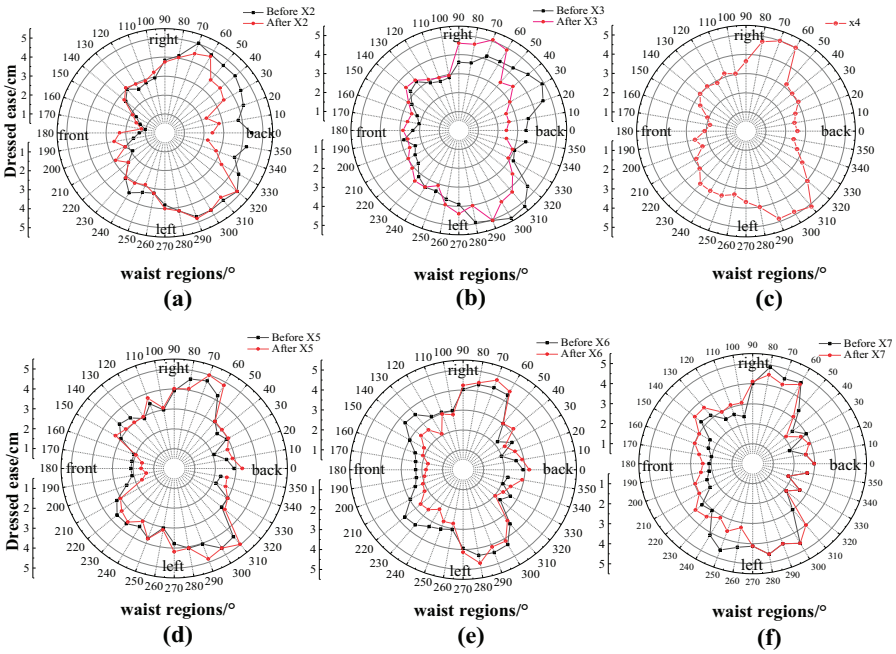

Figure 12. Dressed ease on waist section of modified garment

Waist regions of bust convex angle	X1	X2	X3	X4	X5	X6	X7
0º (cm)	−0.044	−0.101	−0.016	−0.011	0.075	0.078	0.042
30º (cm)	0.013	0.14	0.073	0.113	0.102	−0.096	−0.145
60º (cm)	−0.141	0.097	0.108	0.028	−0.17	−0.126	0.108
90º (cm)	−0.045	−0.041	0.045	−0.021	0.032	0.154	0.106
120º (cm)	−0.004	−0.027	−0.002	−0.01	−0.102	0.183	0.084
150º (cm)	−0.18	0.19	−0.021	0.008	−0.186	0.103	0.079
180º (cm)	−0.121	0.049	0.084	−0.081	0.151	−0.059	0.045

Table IX. Deviation of modified garments' dressed ease

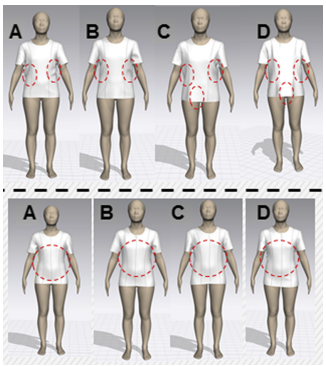

Plate 4. Virtual fitting comparison diagram

Concluding comments

In this study, the dressed ease distributions of difference garments with different bust convex angles and back slope angles were investigated using 3D scanning technique. Based on the influence of bust convex angle on dressed ease distribution at waist section, the 2# garment was modified to have a better fit in the waist of human body with different bust types.

The experiment results showed bust convex angles that influenced the dressed ease distribution weakly were X1 and X2, X3 to X7 had a great influence on dressed ease distribution. Back slope angles ranging from Y1 to Y3 had a great influence on the dressed ease distribution. With the increase of back slope angles, it has a smaller influence. Dressed ease distribution of 1# garment was not affected by bust convex angles and back slope angles ($r < 0.5$ when waist ease allowance is 6 cm). Bust convex angles and back slope angles were highly correlated with dressed ease distribution when garments were 2# to 5# ($r > 0.8$), indicating that it was affected by both bust convex angles and back slope angles. There was a low correlation between dressed ease distribution in the garments of 6# to 8# and bust convex angles ($r < 0.5$), and a moderate correlation with back slope angle ($r > 0.5$). Garment basic construction can be adjusted by the dressed ease distribution to achieve garment fitness. For this reason, the surface convex angle of human body should be evaluated.

This paper provides a technical method for the study of the dressed ease between human body and garment; the results can provide guidance for the optimization of different body shapes. At the same time, the application of dressed ease data to 3D virtual fitting can greatly improve the authenticity of virtual simulation effect.

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